SUPPLY CHAIN ANALYTICS

Sales, Profitability & Delivery Performance Analysis

Python (Pandas, NumPy, Matplotlib, Seaborn) + Google Colab + Power BI

Document Item	Details
Project Type	End-to-End Data Analytics Capstone Project
Dataset	DataCo Supply Chain Dataset
Primary Analysis	Supply Chain, Sales, Profitability and Delivery Performance
Python Environment	Google Colab
Visualization	Matplotlib, Seaborn and Power BI
Primary BI Output	Interactive Supply Chain Management Dashboard
Audience	Business managers and non-technical stakeholders

1. Project Overview

This project analyzes historical supply-chain transaction data to understand business performance, profitability, product performance and delivery efficiency. The objective is to convert raw operational data into clear business insights and practical recommendations.

The project follows a complete analytics lifecycle: business problem definition, data understanding, data cleaning, feature engineering, exploratory data analysis, business insight generation and interactive Power BI reporting.

1.1 Project Title

Supply Chain Analytics: Analyzing Sales, Profitability & Delivery Performance to Identify Operational Improvement Opportunities

1.2 Business Context

A supply-chain/e-commerce organization manages orders across multiple products, customers, markets, regions and shipping modes. Poor delivery performance, low-margin products, excessive discounts or regional performance differences can affect customer satisfaction and business profitability.

2. Business Problem Statement

Management needs a clear view of where the supply chain is performing well and where improvement is required. The organization has historical order, sales, product, customer and logistics data, but raw transactional data does not directly show the most important operational problems.

The project will identify patterns in sales, profit, discounts, products, customer segments, markets and delivery performance, and communicate those findings through an easy-to-understand Power BI dashboard.

2.1 Core Management Questions

- Are we generating healthy sales and profit?
- Are orders being delivered efficiently and on time?
- Which products, markets, regions and shipping modes require management attention?
- Where are the largest opportunities for operational or profitability improvement?

3. Project Objectives

- Understand the structure and quality of the dataset.
- Clean and prepare the data using Python and Pandas.
- Create meaningful business features using Pandas and NumPy.
- Perform exploratory data analysis using Matplotlib and Seaborn.
- Analyze sales, profitability, product, customer and delivery performance.
- Identify operational problems and important business patterns.
- Create an order-level analytical view where required to avoid incorrect KPI calculations.
- Create an interactive Power BI dashboard for non-technical stakeholders.
- Translate analytical findings into business recommendations.

4. Dataset Understanding

The DataCo Supply Chain dataset contains transaction-level supply-chain records covering orders, customers, products, sales, profitability, markets and shipping information.

Data Area	Examples of Fields	Business Use
Orders	Order Id, Order Item Id, Order Status, Order Date	Order volume and trends
Customers	Customer Id, Customer Segment, Country, State	Customer and geographic analysis
Products	Product Name, Category Name, Product Price	Product and category performance
Sales	Sales, Quantity, Discount	Revenue and demand analysis
Profitability	Order Profit Per Order, Profit Ratio	Profit and margin analysis
Logistics	Shipping Mode, Actual/Scheduled Shipping Days	Delivery performance
Geography	Market, Region, Country, State, City	Regional performance

4.1 Important Granularity Consideration

The dataset contains approximately 180,519 order-item records and approximately 65,752 unique orders. One order can contain multiple item rows. Therefore, line-item metrics and order-level metrics must be handled separately.

- Item-level analysis: product, category, quantity, discount and product performance.
- Order-level analysis: order value, total order profit, customer, shipping and delivery performance.
- Use Order Id when calculating unique orders and order-level KPIs.

5. Project Methodology

1. Define the business problem and key management questions.
2. Inspect structure, data types, missing values and duplicates.
3. Clean the dataset without unnecessarily deleting useful records.
4. Create analytical features such as delivery delay and profit margin.
5. Perform EDA by sales, profitability, delivery, customer, product and geography.
6. Convert findings into business problems and recommendations.
7. Prepare a clean analytical dataset for Power BI.
8. Build DAX measures, visuals, filters and drill-downs.
9. Validate KPIs between Python and Power BI.
10. Present the story as What is happening → Where → Why it may be happening → What action to consider.

6. Python / Google Colab Workflow

	Purpose	Main Activities
01_Data_Understanding	Understand raw data	Shape, columns, data types, descriptive statistics, unique values and granularity
02_Data_Cleaning	Prepare reliable data	Missing values, duplicates, irrelevant fields, dates, types and validation
03_Feature_Engineering	Create business metrics	Delivery delay, delivery performance, profit margin, discount amount and order-level data
04_EDA	Discover patterns	Sales, profit, delivery, product, customer, market and geographic analysis
05_Business_Insights	Document findings	Problem, evidence, possible cause,

		impact and recommended action

6.1 Python Libraries

- Pandas — loading, cleaning, grouping, aggregation and transformation.
- NumPy — numerical calculations and feature engineering.
- Matplotlib — core analytical visualizations.
- Seaborn — statistical and relationship-based visualizations.
- Google Colab — development and execution environment.

7. Data Cleaning Plan

Area	Check	Treatment
Duplicate records	Check complete duplicate rows	Remove only confirmed duplicates
Missing values	Profile missingness by column	Handle based on business importance
Completely missing fields	Identify columns with no useful values	Remove if not required
Customer fields	Review missing customer attributes	Use Unknown or retain only if needed
Date fields	Convert order/shipping dates	Create proper datetime columns
Numeric fields	Check types and invalid values	Convert and validate
Categorical fields	Check inconsistent categories	Standardize where necessary
Sensitive/irrelevant fields	Identify fields not required	Exclude from analytical model

8. Feature Engineering Plan

Feature	Logic	Business Meaning
Delivery Delay	Actual shipping days − scheduled shipping days	Early, on-time or late delivery
Delivery Performance	Early / On Time / Delayed	Simple business classification
Profit Margin %	Profit ÷ Sales × 100	Profitability relative to sales
Discount Amount	Price × Quantity × Discount Rate	Estimated monetary discount
Order Value	Sum of sales by Order Id	Total order value
Order Profit	Sum of profit by Order Id	Total order profit
Order-level Margin	Order Profit ÷ Order Value × 100	Order profitability
Year / Month / Quarter	Derived from Order Date	Time trend analysis

9. Exploratory Data Analysis (EDA)

9.1 Business Overview

- Total unique orders
- Total sales
- Total profit
- Average order value
- Profit margin
- Late delivery percentage

9.2 Sales Analysis

- Monthly/yearly sales trend
- Sales by market
- Sales by customer segment

- Sales by category
- Top products by sales

9.3 Profitability Analysis

- Profit by market/category
- Top and bottom products by profit
- Profit margin distribution
- High-sales/low-profit products
- Discount versus profitability

9.4 Delivery & Logistics

- Late delivery percentage
- Delivery by shipping mode
- Delivery by market/region
- Actual versus scheduled days
- Categories with higher delivery risk

9.5 Customer Analysis

- Sales by customer segment
- Profit by customer segment
- Order activity by market
- Order value by segment

9.6 Geographic Analysis

- Sales by market
- Profit by region
- Orders by country/state
- Late delivery by geography

10. Key Business Analysis Questions

- How are sales and profit changing over time?
- Which markets generate the most sales and profit?
- Which categories and products contribute most to revenue?
- Which products have high sales but low profitability?
- Does higher discounting appear to be associated with lower profit margin?
- Which shipping modes have the highest late-delivery rates?
- Which markets or regions have the most delivery problems?
- How different are actual shipping days from scheduled shipping days?
- Which customer segments contribute most to sales and profit?
- Where should management focus first based on poor profitability or delivery performance?

11. Business Insight Framework

Every major finding should follow this structure instead of simply displaying a chart.

Step	Question
Problem	What is happening?

Evidence	What does the data show?
Possible Cause	What factors are associated with the issue?
Business Impact	Why does it matter?
Recommended Action	What should management investigate or improve?

Use terms such as 'associated with', 'related to' or 'may indicate' when the data shows correlation or patterns but does not establish causation.

12. Power BI Dashboard Plan

Page 1 — Executive Overview

Purpose: Give management a 30-second understanding of overall performance.

- Total Sales
- Total Orders
- Profit Margin
- Late Delivery
- Average Order Value
- Sales trend
- Profit by Category

Page 2 — Sales & Profitability

Purpose: Understand revenue and profitability drivers.

- Sales trend
- Sales by category
- Profit by category
- Top products
- Bottom products
- Sales vs Profit
- Discount vs Profit
- Product profitability table

Page 3 — Delivery & Logistics

Purpose: Identify where delivery performance is weak.

- Late Delivery %
- Average Delivery Days
- Average Delay
- Late Delivery by Shipping Mode
- Late Delivery by Months
- Actual vs Scheduled Shipping Days

13. Expected Business Recommendation Areas

Final recommendations must be based on actual findings. Potential areas include:

- Review shipping-mode allocation and delivery commitments in areas with consistently high delays.
- Investigate high-sales products with weak profit margins for pricing or discount optimization.
- Review markets or regions with weak profitability despite strong sales.
- Monitor high-volume products closely because disruptions can have larger business impact.
- Use customer-segment performance to prioritize commercial and service strategies.
- Create recurring monitoring of sales, profit and delivery KPIs through Power BI.

14. Making the Project Understandable to Non-Technical Users

- Use business terms instead of technical terms wherever possible.
- Every chart should answer a specific business question.
- Show the most important KPI first and detailed analysis afterward.
- Use clear titles such as 'Late Deliveries by Shipping Mode'.
- Add short insight statements beside important visuals.
- Explain what management can do with the finding.
- Avoid unnecessary statistical terminology on the executive dashboard.
- Use consistent units, dates and percentage definitions.

15. Final Project Deliverables

Deliverable	Output
Python notebooks	Data understanding, cleaning, feature engineering and EDA
Clean dataset	CSV prepared for BI analysis
Order-level dataset	Aggregated order-level analytical view
Power BI report	Interactive 5-page management dashboard
Project documentation	Business problem, methodology, findings and recommendations
README	Project overview, tools, workflow and key insights
Presentation	Business-focused explanation of problem, analysis and recommendations

16. Final Project Story

- What is happening? — Understand overall sales, profit and delivery performance.
- Where is it happening? — Identify the markets, regions, products and shipping modes involved.
- What factors are associated with the problem? — Analyze discounts, product mix, shipping mode and geography.
- Why does it matter? — Connect the finding to profitability, customer experience or operational efficiency.
- What should management do? — Recommend practical areas for further investigation or improvement.

17. Success Criteria

- The dataset is cleaned and validated with clear treatment decisions.
- Major KPIs are calculated at the appropriate data grain.
- EDA identifies meaningful patterns rather than only producing visualizations.

- At least 5–10 strong business insights are documented with evidence.
- Power BI allows a non-technical stakeholder to understand key findings quickly.
- Recommendations are directly connected to analytical findings.
- Python and Power BI results are consistent for major KPIs.